

ResultGuard

A Browser-Based Cryptographic Framework for Academic Result Authentication and Tamper Detection

1. Abstract

The increasing use of digital academic records has created a growing need for reliable methods to verify the authenticity and integrity of academic results. Traditional verification methods often depend on manual document inspection or direct communication with the issuing institution, which can be time-consuming and inefficient.

This project proposes ResultGuard, a browser-based cryptographic framework for academic result authentication and tamper detection. The system will use SHA-256 for cryptographic hashing and RSA-2048 with RSA-PSS for digital signatures. An authorized user will generate an academic result and digitally sign it using a private key. An authorized verifier will then use the corresponding public key to verify the result. Any modification to the signed information should cause the verification process to fail.

The project will be developed as a lightweight browser-based prototype using HTML5, Tailwind CSS, Vanilla JavaScript, and the Web Crypto API. The proposed system aims to demonstrate a practical application of cryptographic techniques for improving the integrity, authenticity, and reliability of digital academic records.

Keywords: Academic Result Authentication, Digital Signature, SHA-256, RSA-2048, RSA-PSS, Cryptography, Tamper Detection.

2. Introduction

Academic results are important records used for university admission, employment, scholarships, internships, and other official purposes. With the increasing adoption of digital systems, academic results are now frequently generated, stored, and distributed electronically.

Although digital records improve accessibility and reduce dependence on physical documents, they also introduce concerns regarding authenticity and integrity. A digital academic document may potentially be copied or modified after it has been issued. Traditional verification processes may require organizations to manually inspect documents or contact the issuing institution.

Cryptographic techniques can provide a reliable approach to this problem. Cryptographic hashing can generate a unique representation of academic result data, while digital signatures can provide a mechanism for verifying that the data was signed by an authorized entity and has not been modified.

Based on these principles, ResultGuard proposes a browser-based academic result authentication system that enables results to be digitally signed and independently verified.

3. Problem Statement

The authenticity of digitally distributed academic results is an important concern for educational institutions, students, employers, and other organizations. Traditional verification methods may rely on visual inspection, manual communication, or direct confirmation from the issuing institution.

These methods can require additional time and administrative effort and may not provide direct cryptographic evidence that a document has remained unchanged.

For example, information such as a student's name, grade, CGPA, or result status could potentially be modified after a result has been issued. Detecting such modifications through visual inspection alone may be difficult.

Therefore, there is a need for a simple mechanism that can automatically verify whether an academic result is authentic and whether its contents have been altered after issuance. ResultGuard aims to address this problem through cryptographic hashing and digital signatures.

4. Aim & Objectives

4.1 Aim

The main aim of this project is to develop a browser-based academic result authentication and verification system that uses cryptographic techniques to verify the integrity and authenticity of digital academic records and detect unauthorized modifications.

4.2 Objectives

- Develop a browser-based academic result authentication system.
- Generate structured digital academic results.
- Apply SHA-256 hashing to academic result data.
- Implement RSA-2048 public-key cryptography.
- Use RSA-PSS for digital signatures.
- Digitally sign academic results using an authorized private key.
- Verify signed results using the corresponding public key.
- Detect unauthorized modifications to academic records.
- Provide a simple and user-friendly verification interface.
- Demonstrate the practical application of cryptography in academic record security.

5. Project Significance

The proposed project is significant because it demonstrates how modern cryptographic techniques can be applied to a practical problem in the education sector.

Digital academic records are increasingly used for admission, employment, scholarships, internships, and other official purposes. A reliable verification mechanism can reduce dependence on manual document checking and provide stronger evidence of record integrity.

ResultGuard will demonstrate how digital signatures can establish a verifiable relationship between an academic result and its authorized signing key. This can help organizations determine whether submitted academic information has remained unchanged after issuance.

The project also provides practical experience in applying hashing, public-key cryptography, and digital signatures to a real-world software system.

6. Proposed Methodology

The proposed methodology follows a structured development process beginning with problem identification and requirement analysis. The system architecture and user workflow will then be designed before implementing the cryptographic components.

The implementation will focus on SHA-256 hashing, RSA-2048 key generation, and RSA-PSS digital signatures. Academic results will be generated and digitally signed before being made available for verification.

Finally, the verification and tamper-detection process will be evaluated through functional testing to confirm that authentic results are accepted and modified results are rejected.

Proposed ResultGuard Methodology

Structured development process from problem identification to final prototype

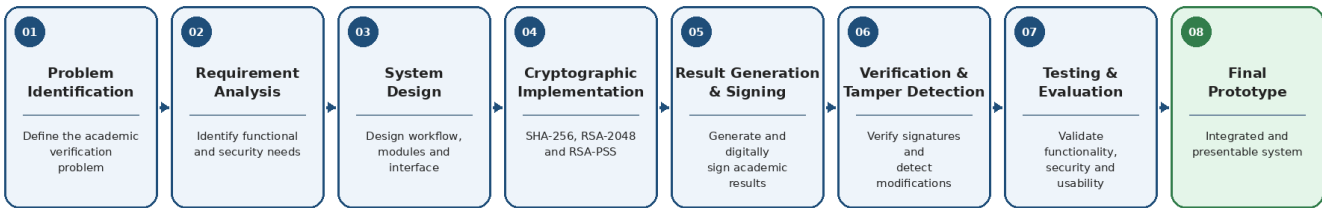


Figure 1. Proposed ResultGuard Methodology

7. System Architecture

The proposed system consists of result generation, cryptographic processing, result sharing, and verification stages. An authorized user generates the academic result, which is hashed using SHA-256 and digitally signed using RSA-PSS with the private key.

During verification, the submitted result is checked using the corresponding public key. If the signature is valid and the result remains unchanged, the system reports an authentic result. If the result has been modified or the signature is invalid, the system reports an invalid or tampered result.

Proposed ResultGuard System Architecture

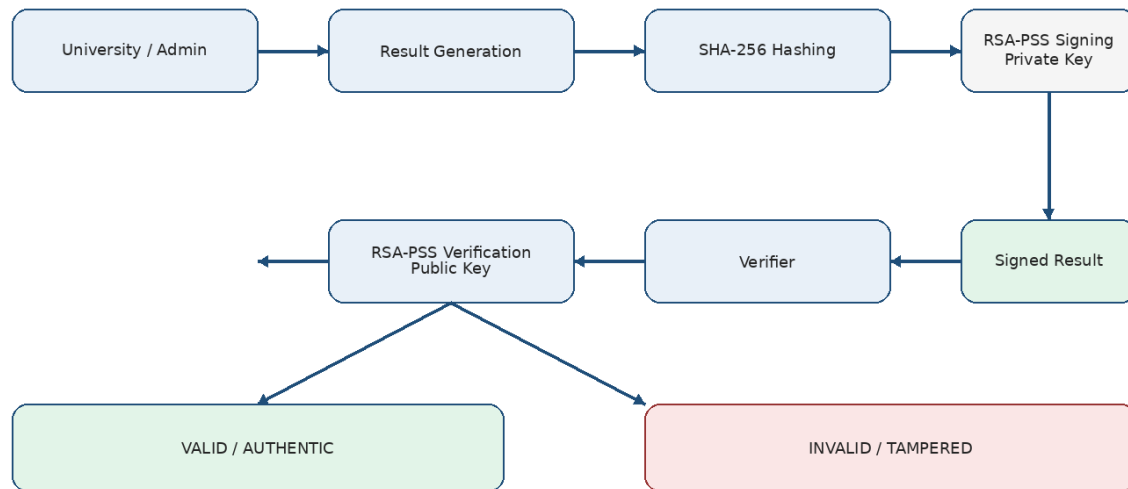


Figure 2. Proposed ResultGuard System Architecture

8. Technology Stack

Technology	Purpose
HTML5	Application structure
Tailwind CSS	User interface and responsive design
Vanilla JavaScript	Application logic and data processing
Web Crypto API	Browser-based cryptographic operations
SHA-256	Cryptographic hashing
RSA-2048	Public-key cryptography
RSA-PSS	Digital signature
Git/GitHub	Version control and project management

9. Team Members

Team Member	Role
Hasibul Hoque Probal	Project Lead
Mosaddek Hossain Mehedi	Software Developer
Sultana Yeasmin Suma	UI/UX Designer
Azizur Rahman Arzu	Testing & Documentation

10. Scope

The project focuses on developing a proof-of-concept academic result authentication and verification system.

Included in Scope

- Academic result generation
- Student and academic information processing
- CGPA calculation
- Input validation
- SHA-256 hashing
- RSA-2048 key generation
- RSA-PSS digital signatures
- Result verification
- Tamper detection
- Responsive user interface
- Cryptographic verification workflow

Outside the Initial Scope

- Production university database
- Centralized authentication server
- Blockchain infrastructure
- Certificate Authority
- Hardware Security Module
- Large-scale institutional deployment
- Production-grade private-key infrastructure

11. Expected Outcomes

At the end of the project, the team expects to deliver a functional browser-based prototype capable of generating, digitally signing, and verifying academic results.

- Generate structured academic results.
- Create SHA-256 cryptographic hashes.
- Generate RSA-2048 key pairs.
- Create RSA-PSS digital signatures.
- Verify digitally signed results.
- Detect unauthorized modifications.
- Display clear verification results.
- Demonstrate the practical application of cryptography in academic record protection.

The project will also provide a practical demonstration of how digital signatures can improve trust in electronically distributed academic records.

12. Future Scope

The proposed prototype can be extended into a more comprehensive academic credential verification platform.

- Secure backend-based digital signing
- University authentication
- Digital certificates
- QR-code-based verification
- Online verification portal
- Public verification API
- Key revocation and rotation
- Audit logging
- Cloud deployment
- Multi-university support
- Digital credential wallets
- Institutional database integration

These improvements could transform ResultGuard from an academic prototype into a scalable digital academic credential verification platform.

13. Conclusion

ResultGuard proposes a practical approach to academic result authentication using modern cryptographic techniques. By combining SHA-256 hashing, RSA-2048, and RSA-PSS digital signatures, the system will provide a mechanism for verifying the integrity and authenticity of digitally issued academic results.

The proposed system focuses on developing a lightweight, browser-based proof-of-concept that demonstrates the complete process of result generation, cryptographic signing, verification, and tamper detection.

Although the initial implementation is limited to a prototype environment, the project demonstrates how cryptographic techniques can be applied to a real-world academic problem. The resulting system can serve as a foundation for future development of a more secure, scalable, and institution-ready digital academic credential verification platform.

14. Result Verification and Tamper Detection

The verification stage compares the submitted result with its cryptographic signature. An unchanged result should produce successful signature verification, while a modified result should fail verification and be identified as potentially tampered.

Result Verification and Tamper Detection

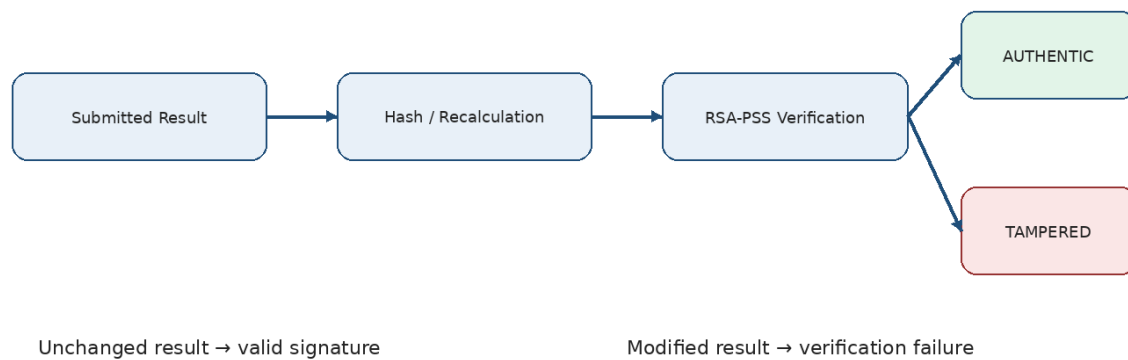


Figure 3. Result Verification and Tamper Detection

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